

DATA STRUCTURES - Examination 2023

Total Marks: 70

SECTION A

1.

Attempt all questions in brief.

- (a) Define array and linked list.
- (b) What is a stack? Give two applications.
- (c) State the time complexity of binary search.
- (d) Define a queue and its types.
- (e) What is a binary search tree?
- (f) Differentiate between BFS and DFS.
- (g) What is hashing?

SECTION B

2.

Attempt any three of the following:

- (a) Explain the difference between array and linked list with examples.
- (b) Explain push and pop operations of a stack with an algorithm.
- (c) Describe tree traversal techniques with examples.
- (d) Explain hashing and collision resolution techniques.
- (e) Explain quick sort and derive its average case complexity.

SECTION C

3.

Attempt any one part of the following:

- (a) Explain insertion and deletion in a binary search tree.
- (b) Explain Dijkstra shortest path algorithm with an example.

4.

Attempt any one part of the following:

- (a) Write the binary search algorithm and explain its complexity.
- (b) Explain AVL tree rotations with examples.